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CO3050 - Web Programming (Lab)

Report

Individual Project

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1 Introduction

This report details the development of the **BookStore** application, a full-stack e-commerce platform designed to facilitate the online sale of books. The project aims to provide a seamless user experience for customers while offering robust administrative tools for inventory and location management.

2 System Architecture

The application follows a decoupled architecture separating the user interface from the business logic and data storage.

2.1 Frontend

The frontend is built using **React 18** and **Vite**.

- **Framework:** React with Functional Components and Hooks.
- **Routing:** Client-side routing managed by `react-router-dom`.
- **State Management:** Context API (`AuthContext`, `CartContext`).
- **Styling:** **Tailwind CSS** for a utility-first, responsive design.

2.2 Backend

The backend consists of a RESTful API implemented in **PHP**.

- **API Entry Point:** A central `index.php` handling routing through regular expressions.
- **Database Connection:** Managed via **PDO** for security and support for prepared statements.
- **Authentication:** Token-based authentication using Base64 encoded JSON payloads for session validation.



2.3 Security & Configuration

Environment-specific variables (Database credentials, SMTP settings) are stored in a `.env` file. This practice prevents the accidental disclosure of sensitive credentials in the source code. A PHP helper script `env.php` was implemented to load these variables into the server environment.

3 Database Design

The system utilizes a relational database (**MySQL**) with the following primary entities:

1. **Users:** Stores user credentials, roles (admin/customer), and profile data.
2. **Products:** Contains book information including category, title, author, description, price, and cover image paths.
3. **Categories:** Organizes products into logical groupings (e.g., Fiction, Science).
4. **Locations:** Stores physical store or warehouse data (name, address, maps URL).
5. **ProductLocations:** A junction table managing many-to-many relationships between products and locations, including specific stock levels.

4 Features

- **User Authentication:** Secure registration, login, and a password reset flow using Mailtrap for SMTP testing.
- **Product Catalog:** Dynamic browsing with filtering by category and detailed product views.
- **Shopping Cart:** Real-time cart management across the session.
- **Admin Dashboard:** A restricted interface allowing administrators to:
 - Create, Update, and Delete products.
 - Manage multiple store locations and track localized inventory.
 - Process application-wide settings.



5 Installation and Setup

To deploy the project locally:

1. Install Node.js dependencies: `npm install`.
2. Configure the `.env` file based on `.env.example`.
3. Import the database schema into a MySQL server named `online_book_store`.
4. Run the development server: `npm run dev`.

6 Conclusion

The BookStore project successfully integrates modern frontend technologies with a reliable PHP backend. The use of environmental configuration and prepared statements ensures a baseline level of security, while the React-based frontend delivers a high-performance interactive experience for users.